JLPT AI App, Progress Report No. 01

Padauk Lab

Report	Period	Date	Prepared by
No. 01	Oct 9 to Oct 12, 2026	Oct 12, 2026	HP (Backend and Infrastructure)

You can share this file with your AI coding tool as project context.

1. Summary

The team workspace is set up and the backend foundation is built. The backend runs with one command (docker compose up) and answers on /health. This work is in pull request #1, waiting for a teammate's review.

2. Project at a glance

Item	Details
App	GenAI JLPT study app for Burmese speakers (web, iOS, Android)
Repository	github.com/padauk-lab/ai-jlpt-app
Frontend	Expo (React Native)
Backend	FastAPI, Python 3.12
Database	PostgreSQL 16
Local setup	Docker and Docker Compose

3. Completed this period

No.	Task	Owner	Status
1	GitHub organization (padauk-lab), team (core) and repository	HP	Done
2	Branch protection on main (pull request and 1 approval required)	HP	Done
3	Starter folders, README.md, CONTRIBUTING.md, .gitignore, .env.example	HP	Done
4	Docker Desktop installed and tested	HP	Done
5	FastAPI backend with GET /health returning {"status":"ok"}	HP	In review (PR #1)
6	Backend Dockerfile (Python 3.12 slim, non-root user)	HP	In review (PR #1)
7	docker-compose.yml (PostgreSQL 16 and backend, with health check)	HP	In review (PR #1)
8	Locked Python dependencies (requirements.in and requirements.txt)	HP	In review (PR #1)
9	Team roles document and project brief	HP	Done

4. Folder ownership

Folder	What goes inside	Owner
frontend/	Screens and UI	LMHP
backend/app/routers/	API endpoints	HP

Folder	What goes inside	Owner
backend/app/models/	Database tables	HP (ANK after Dec 6)
backend/app/services/	Study plan and business logic	HP
backend/app/ai/	AI code	SWYH
data/	Vocabulary, grammar, proverbs, mock test templates	ANK
docs/	API contract and diagrams	Everyone

5. How to run the app (after PR #1 is merged)

1. Install Docker Desktop, and turn on WSL integration if you use WSL
2. `git checkout main` and `git pull`
3. `cp .env.example .env`, then set your own POSTGRES_PASSWORD (also inside DATABASE_URL)
4. `docker compose up --build`
5. Open http://localhost:8000/health and http://localhost:8000/docs
6. Stop with Ctrl+C, then `docker compose down`

6. Rules

1. Never push to main. Create a branch (feature/name or fix/name), push it, open a pull request
2. Every pull request needs 1 approval from a teammate
3. Never commit .env, API keys or passwords
4. Stay inside your own folder, ask before changing someone else's area
5. Python libraries: add to backend/requirements.in, then regenerate requirements.txt with exact versions
6. Using AI is fine, but you must be able to explain everything you commit

7. Next tasks (Week 1, Oct 12 to 18)

No.	Task	Owner	Depends on
1	Review and approve PR #1	Any teammate	Nothing
2	Draft the API contract in docs/ (endpoints and JSON responses)	HP	Nothing
3	Design system (colors, fonts, logo) and screen sketches	LMHP	Nothing
4	Expo project setup in frontend/	LMHP	Nothing
5	Choose the word list source and write one complete sample unit	ANK	Nothing
6	Plan which words and grammar go into each session and unit	ANK	Task 5
7	Compare LLMs on Burmese and Japanese quality and cost	SWYH	Nothing
8	Start the explanation script using the sample unit format	SWYH	Task 5
9	Team meeting: agree on API contract and content format	Everyone	Tasks 2 and 5

8. After Week 1

No.	Task	Owner	Target
1	Google login and database tables	HP	Weeks 2 to 4
2	All screens built with fake data from the API contract	LMHP	Weeks 2 to 4

No.	Task	Owner	Target
3	AI drafts content, ANK reviews	SWYH and ANK	Weeks 2 to 4
4	First integration: login, dashboard and one unit end to end	Everyone	Week 5 (Nov 9 to 15)
5	Automatic checks on pull requests (GitHub Actions)	HP	Later sprint

9. Open decisions

1. Which LLM to use (SWYH to propose)
2. Which vocabulary and grammar source list to use (ANK to propose)
3. Meeting date for the API contract and content format